

Rafael Augusto do Nascimento

AI, Geospatial Technologies and Public Innovation

I develop data-driven and AI-powered solutions for smart cities, public management and territorial analysis.

My work combines public-sector experience with technologies such as Python, Geographic Information Systems, Large Language Models, Retrieval-Augmented Generation and public data analysis.

Current areas of interest

- Geospatial Artificial Intelligence — GeoAI
- AI agents for spatial analysis
- Smart cities and digital government
- Environmental and territorial intelligence
- Retrieval-Augmented Generation — RAG
- Public data automation and decision support

Featured project

Municipal AI Diagnosis

AI-powered system that analyzes municipal indicators and generates technological maturity assessments using:

- Python
- LangChain
- Google Gemini
- FAISS
- Semantic embeddings
- Pandas and Pydantic
- Automated Word report generation

[View the project](#)

Technologies



Currently developing

A GeoAI framework for creating AI agents capable of accessing geospatial data, performing spatial operations and answering territorial questions through natural language.

Professional focus

Building practical applications of artificial intelligence and geospatial technologies for governments, cities and environmental management.